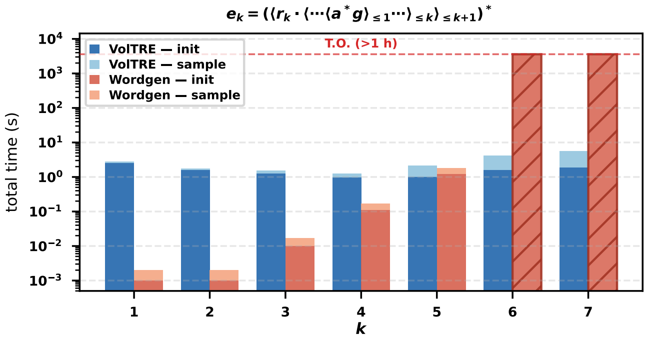

Fig. 1: Total time (preprocessing + 10 samples, log scale, $n = 10$) for the expression family e_k as k varies. Blue: VolTRE (TRE-based). Red: WORDGEN (TA-based). Hatched bars indicate time-out (>1 h); the dashed line marks the cutoff.

PREVIEW: EXPERIMENT 16 — TA VS. TRE SCALABILITY

Scalability on nested request-grant protocols.

We compare VolTRE against the TA-based sampler WORDGEN [wordgen] on a parameterised family that stresses the clock-resource gap between the two formalisms. Consider the following k-level request-grant expression:

$$e_k = \Big(\langle r_k \cdot \langle r_{k-1} \cdot \langle \cdots \langle r_1 \cdot \langle a^* g\rangle_{\le 1}\rangle_{\le 2} \cdots \rangle_{\le k-1}\rangle_{\le k}\rangle_{\le k+1}\Big)^* \tag{1}$$

Each cycle of e_k models a protocol with k request types $r_k, r_{k-1}, \ldots, r_1$ issued in sequence, followed by a grant g within 1 time unit of the final request r_1. The nested constraints enforce cumulative deadlines: the sub-sequence from r_i must complete within $i+1$ time units. An equivalent timed automaton requires k clocks—one per deadline level—so its state space grows exponentially in the number of clocks, posing a scalability challenge for TA-based sampling approaches. VolTRE, by contrast, processes each $\langle \cdot \rangle_{\le i}$ layer algebraically through the slice-volume recursion, with only a polynomial overhead per level.

a) Setup.: We fix word length $n = 10$ and vary $k = 1, \ldots, 7$. For each k we measure (i) the one-time preprocessing time (t_{vol} for VolTRE, t_{split} for WORDGEN) and (ii) total time including 10 samples. WORDGEN was given a budget of 3600 s. VolTRE scales to larger n than shown here; we cut off where WORDGEN fails.

b) Results.: Figure 1 shows total time on a logarithmic scale. VolTRE processes all values of k in well under 15 s, with per-sample cost growing gradually from ≈ 0.2 s at $k{=}1$ to ≈ 5.6 s at $k{=}7$. WORDGEN is faster for small k (total ≈ 1.8 s at $k{=}5$), but fails to terminate within the 1 h budget at $k{=}6$ due to memory exhaustion, and by extension for all $k \ge 6$. This illustrates a structural advantage of the TRE representation: the exponential blowup in the number of clocks is avoided by VolTRE's compositional volume computation, at the cost of rejection sampling for intersections that TA-based methods handle natively.